

SAP CPI Project Documentation

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Scenario: HR_Timesheets_Processing_and_Validation

1.Scenario Overview

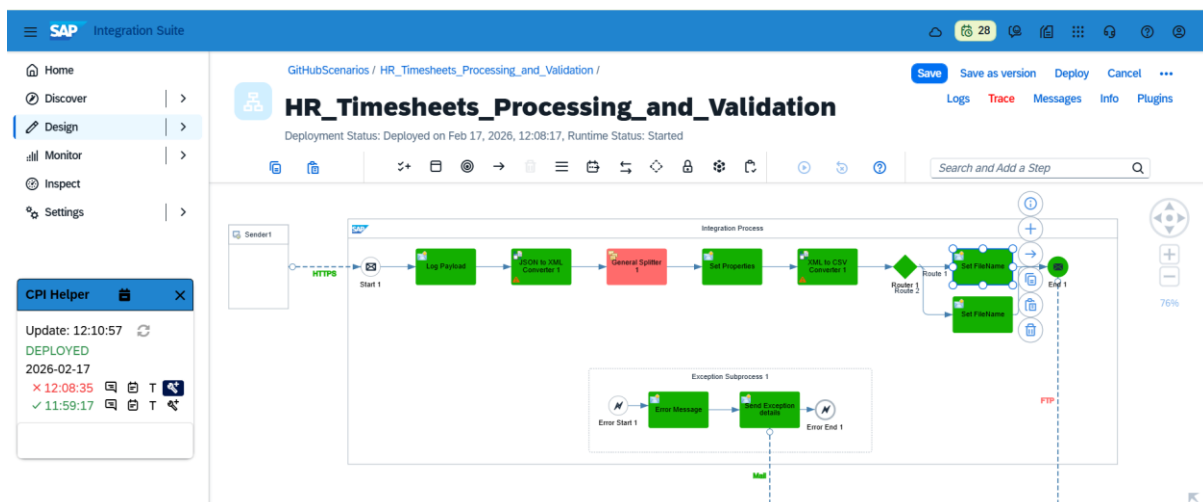
This integration handles employee timesheet submissions from a web application. Timesheet records are validated, converted, and stored in the HR system in CSV format using region-specific folders. It demonstrates the use of HTTP Adapter, Splitter, Filter, Router, Format Conversion, and File Adapter, suitable for automating and validating HR operations.

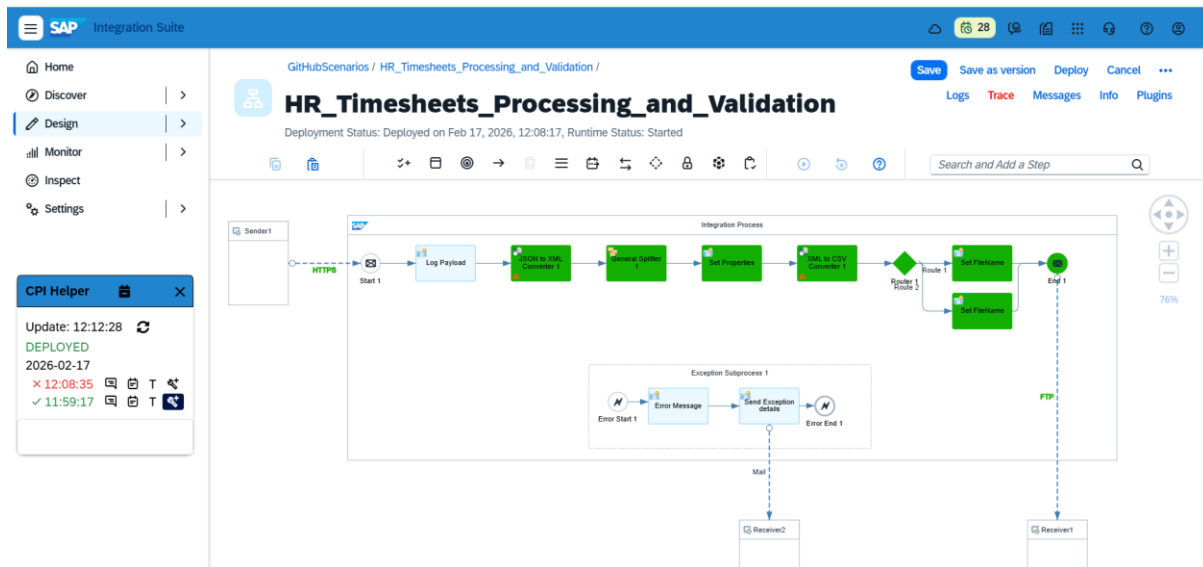
Business Use Case

Employees submit daily timesheets via a **custom web application**. Each timesheet record must be:

- Validated (e.g., working hours ≤ 12)
- Converted into a **CSV format**
- Stored in an HR system using a **File Receiver Adapter**
- Invalid records should be **logged separately** for compliance review

2.Architechture Diagram





3. Integration Flow

Steps

1. HTTP Sender Adapter receives batch JSON data

HTTPS

General **Connection** Conditions

REQUEST PROCESSING

Address:

Authorization:

User Role:

CSRF Protected: ☒

2. JSON to XML Converter prepares structure for processing
3. XML Splitter breaks down entries into individual records

General Splitter

General **Processing**

Expression Type:

XPath Expression:

Grouping:

Streaming: ☒

Parallel Processing: ☐

Stop on Exception: ☐

4. **Content Modifier** adds metadata (e.g., company name, processed timestamp) by **Setting Properties**

The screenshot shows the 'Content Modifier' interface with the 'Exchange Property' tab selected. It displays a table with columns: Action, Name, Source Type, Source Value, Data Type, and Default Value. Five properties are listed:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	EmployeeID	XPath	//EmployeeID	java.lang.String	
Create	Date	XPath	replace(//Date, ',', '')	java.lang.String	
Create	Hours	XPath	//Hours	java.lang.String	
Create	CompanyName	Constant	Infosys		
Create	ProcessedDate	Expression	\$(date:now)		

5. **Router** separates valid and invalid records using exchange property

The screenshot shows the 'Router' interface with the 'Processing' tab selected. It displays a table with columns: Order, Route Name, Condition Expression, and Default Route. Two routes are defined:

Order	Route Name	Condition Expression	Default Route
1	Route 1	`\${property.Hours} <= '12'`	☐
2	Route 2		☒

6. **XML to CSV Converter** formats data
7. **File Receiver Adapter** writes valid CSVs to /hr/valid/, invalid ones to /hr/invalid/

The screenshot shows the 'FTP' interface with the 'Target' tab selected. It displays two sections: 'FILE ACCESS PARAMETERS' and 'CONNECTION PARAMETERS'.

FILE ACCESS PARAMETERS:

- Directory: /hr/\${property.directory}
- File Name: \${property.EmployeeID}_\${property.Date}.csv
- Append Timestamp: ☐

CONNECTION PARAMETERS:

- Address: ftp.drivhq.com
- Proxy Type: Internet
- Encryption: Explicit FTPS
- Credential Name: FTP_Connect
- Timeout (in ms): 10000
- Maximum Reconnect Attempts: 3

8. Sample Input JSON (from Web App)

```
{
  "Timesheets": [
    { "EmployeeID": "E123", "Date": "2025-07-16", "Hours": 9 },
```

```
{ "EmployeeID": "E456", "Date": "2025-07-16", "Hours": 14 }
]
}
```

9. Converted XML (after JSON to XML)

<Timesheets>

<Entry>

<EmployeeID>E123</EmployeeID>

<Date>2025-07-16</Date>

<Hours>9</Hours>

</Entry>

<Entry>

<EmployeeID>E456</EmployeeID>

<Date>2025-07-16</Date>

<Hours>14</Hours>

</Entry>

</Timesheets>

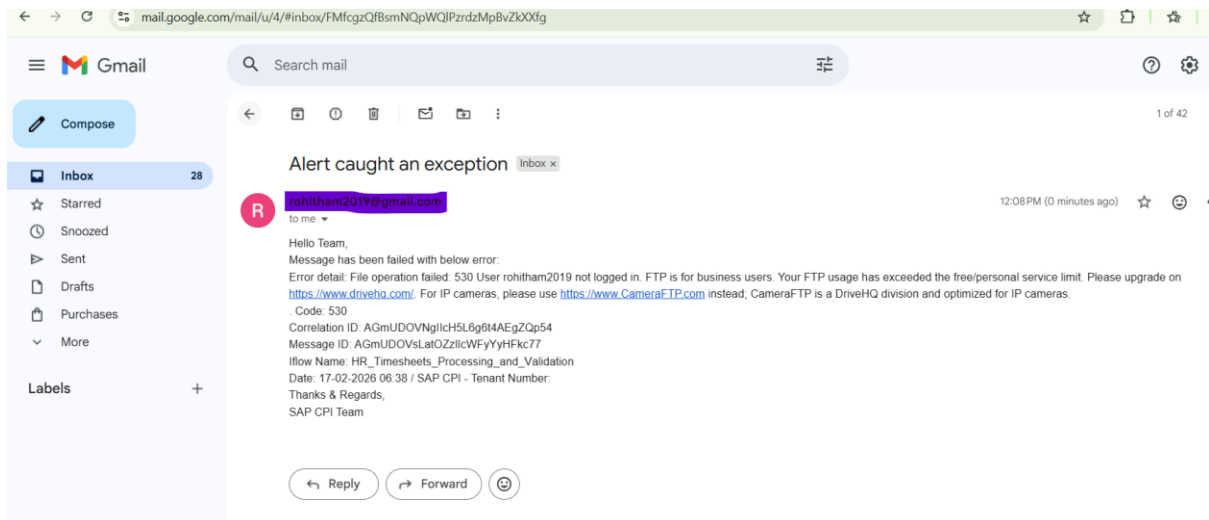
The screenshot shows the DriveHQ web interface. The top navigation bar includes the DriveHQ logo and links for Features, Software, and Pricing. Below the navigation bar is a toolbar with various actions like Up, Upload, Download, Zip Download, Share, Publish, New Folder, New File, Copy, Cut, Delete, and Paste. The main area displays a file explorer view of a folder path: \\Recycle Bin\\2026-02-16\\hr_213522_DeletedItem\\invalid\\. The left sidebar shows the DriveHQ Root and My Storage, with a tree view of folders including ALL_FILES, DriveHQData, Favorites, Invoice, My Documents, My Pictures, Order, Others, Recycle Bin, 2026-02-16, hr_213522_DeletedItem, invalid, valid, hr_223803_DeletedItem, wwwhome, DriveHQShare, webmaster, and My Profile. The right pane shows a table of files in the selected folder.

Name	Action	Create Time	Modify Time	Size
E123_20250716.csv		2/17/2026 11:00:20 AM	2/17/2026 11:00:19 AM	19 B

Files stored in HR system

10. Exception Handling

- Any transformation or file-write error is:
 - Alert sent via **Gmail Adapter** with employee ID and error details



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Iflow Package



HR_Timesheets_Processing_and_Validation.zip